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The *Ice Dart*: Landing on Steep Ice Using Spiny Feet, Dissipative Landing Gear, and Reverse Thrust

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(Regular Article)

ABSTRACT This article presents a strategy that enables small multirotors to reliably land on steep and irregular ice surfaces, such as those found on icebergs and glaciers. The proposed system integrates three key components: 1) ice-gripping feet equipped with opposing, passively retractable spines that generate anisotropic shear forces for both uphill- and downhill-facing landings; 2) a friction-based dissipative landing gear that absorbs kinetic energy and increases spine penetration by preventing rebound; and 3) reverse thrust (RT) at touchdown to increase normal forces and substantially increase shear forces. A detailed study of ice–spine interaction guided the selection of two distinct sets of spine properties for the downhill and uphill spines, each to meet a distinct function, respectively: 1) aggressive braking on steep ice to avoid holes, crevasses, or ice protrusions, and 2) easy spine re-engagement on steep slopes under low normal loads to prevent slipping after landing. A custom quadrotor drone, the *Ice Dart*, was designed for high power and maneuverability in strong winds, enabling validation of the strategy on real icebergs and glaciers. Laboratory drop tests of the *Ice Dart* demonstrated successful landings at descent speeds of 1–3 m/s on ice slopes of up to 60°, while confirming the advantages of the specialized feet. Field trials in Iceland demonstrated the performance on diverse ice types, iceberg sizes and shapes, achieving 24 successful landings on icebergs and glaciers, at inclinations of up to 58° and heading offsets of up to 45°, in winds of up to 30 km/h, and achieving a 100% success rate.

INDEX TERMS Drone, friction shock absorber (FSA), ice, landing, landing gear, multirotor, quadrotor, reverse thrust (RT), spine, uninhabited aerial vehicle (UAV).

I. INTRODUCTION

UNINHABITED aerial vehicles (UAVs) are increasingly used in remote and hazardous environments for monitoring, data collection, and surveillance across various industries. In glaciology and cryology, UAVs serve as versatile remote-sensing platforms for glacier mapping, photogrammetric 3-D reconstruction of icebergs, and long-term tracking of iceberg movement. Despite significant advances in sensors, propulsion, and control systems, UAV landing capabilities remain limited, especially in cryospheric environments. Enabling UAVs to land securely on inclined ice surfaces would expand their applications, such as deploying drone-trackers on icebergs to monitor drift and prevent collisions with offshore installations, safely deploying sensors on glaciers, and collecting in situ ice samples. More broadly, reliable landings on steep, low-friction surfaces would also improve operations beyond icy environments, allowing safe touchdowns on uneven terrain, rocky hills, and rooftops. The present work introduces a landing system that integrates specialized spiny feet, friction shock absorbers (FSAs), and reverse thrust (RT) to facilitate secure

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FIGURE 1. ***Ice Dart*** **landed on the ice facade of the Fjallsjökull glacier (Iceland), right at the glacier's edge with the water.**

landings on steep, slippery ice surfaces inclined up to 60° and at impact speeds up to 3 m/s, as depicted in Fig. 1.

A. STEEP ICE LANDING CHALLENGES

Landing a drone on steep ice is difficult for two main reasons. First, the coefficient of friction (CoF) between the drone's feet and ice is very low. For instance, the CoF is around 0.15 for rubber materials and only 0.06 for 3-D-printed plastic feet. This means the drone would start slipping at slope angles of 9° and 3°, respectively. Second, at high slope angles, the normal force on the feet quickly decreases (e.g., only 50% of the gravity force acts normal to the surface on a 60° slope), making it much harder to generate friction forces to slow down the drone during landing or even maintain its position on the ice.

Numerous studies have explored the use of spines and microspines to grip or climb hard, vertical, low-friction surfaces by engaging surface asperities to generate shear forces. Various implementations have shown that load-sharing microspine arrays are effective for gripping hard rough materials such as concrete, stucco, brick, and masonry [1], [2], [3], [4]. Other research introduced a hybrid rubber–spine composite to increase friction on different surfaces, utilizing rubber on smooth surfaces and nonpenetrating spines on wet but rough hard surfaces [5]. However, unlike hard, nondeformable materials, ice is more easily penetrable, allowing spines to create their own anchoring asperities rather than relying on existing ones. Alternative methods, such as gecko-inspired dry adhesives [6] and suction cups [7], have been proposed to cling to smooth surfaces, but they require large contact areas and are likely to exhibit reduced performance on ice. Few studies have explored the use of spines for adhering to ice and expanding the landing envelope of small UAVs.

Icebergs and glaciers also display highly irregular and unpredictable shapes at the macroscale, with uneven surface profiles, holes, crevasses, and ice protrusions, all of which complicate the process of locating suitable landing sites [8]. Cryospheric and maritime environments also expose UAVs to strong winds and gusts, especially near large ice formations, which can have harmful aerodynamic effects such as causing sudden changes in velocity during landing [9]. Moreover, since these sites are often completely inaccessible, limiting the installation of landing aids and surface preparation for landing or catching, the aerial platform bears full responsibility for safe touchdowns. To mitigate the consequences of failed landings, several studies have therefore opted to deploy only a limited set of sensors rather than full UAVs or complex instruments, either by dropping trackers from UAV tethers [9], manually releasing them from helicopters [10], or dropping them from boats [11]. While most of these studies used some form of nonspecialized spikes to better grip the ice (e.g., wood screws passing through a board or carpet), they still emphasized the importance of finding flat, near-horizontal landing spots. Although effective on large tabular icebergs, these methods limit the range of iceberg shapes and sizes that can be targeted. Furthermore, since UAVs already carry many of the sensors used in trackers, enabling them to safely land on steep, irregular ice surfaces would expand their potential in glaciological fieldwork, facilitating sensor retrieval and redeployment.

Achieving reliable landings on ice presents challenges that also apply to space exploration, where landers must touch down on unpredictable, uneven, and often icy terrains such as comets or moons. These systems often utilize barbed spikes and deployable harpoons, with thrusters to limit rebounds [12], [13], [14]. Failures of the landing system have included large rebounds in low gravity [14], fallen over landers due to the terrain [15], and broken legs at impact [16]. These incidents highlight the importance of robust landing technologies for both terrestrial and extraterrestrial missions.

Even on flat, nonslippery terrain, commercial UAVs face numerous challenges when landing at high impact speeds, which can promote spine engagement during landing, or on

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inclined surfaces. For example, ArduPilot [17], a widely used open-source autopilot, limits the default descent rate to 0.5 m/s, highlighting the limitations of rigid landing gear that act like springs and often lead to bouncing, sliding, or tipping over upon impact. To address these issues, various strategies have been proposed. Corrugated-shell shock absorbers have been used to withstand impacts up to 2 m/s [18], while actuated-leg designs can adapt to slopes of 20°–30° [19], [20] and achieve stable landings on cluttered terrain [21]. Other approaches have integrated lateral rotors to enable preinclined, low-speed approaches [22]. Although effective within their tested ranges, these solutions typically add substantial weight penalties and remain insufficient on their own for reliable landings on steep ice.

Previous work by the authors showed that combining energy-dissipative landing gear with RT enables small multirotors to land on rooftops up to 60° [23] and on various vehicles at speeds of up to 110 km/h [24]. In both cases, the constant-force suspension, which is comprised of FSAs, allows landings at up to 3 m/s by absorbing the drone's kinetic energy upon impact, while RT activated at contact increases the drone's friction with the ground. However, landing on ice poses very distinct challenges compared to engineered surfaces like rooftops. Shingle rooftops, for example, have a CoF at least seven times higher, feature large, flat, uniform surfaces, and are usually less exposed to strong winds. Therefore, landing on steep ice requires addressing a new set of challenges.

B. BALANCING THE TRADEOFFS

Building on previous work utilizing FSAs and RT, this work introduces a novel spiny foot design that employs specialized spines to enable small drones to land on inclined, uneven, and low-CoF surfaces such as icebergs and glaciers. For successful landings, the foot design must navigate many tradeoffs. First, to prevent slipping, quick engagement is necessary. However, to allow the FSAs to dissipate landing energy and prevent rebounds, the landing gear legs must be able to rotate freely, which is only possible if the feet can slide on the ice before penetrating it. This rules out simple spikes that engage immediately upon contact. Second, the spines must be strong enough to generate large shear forces during landing to slow the drone down, but small enough to re-engage quickly under limited normal forces on inclined surfaces. This self-reengaging ability is vital for keeping the drone perched safely for extended periods, as ice can crack or melt after landing. Third, deep spine penetration into the ice could help generate high shear forces, while barbed spines could prevent rebound, but both might also hinder the drone's ability to take off easily. Finally, the dynamics of landing and braking on sloped surfaces create different loads on the uphill and downhill feet, which could be exploited to increase successful landings. That being said, specialized feet could require the UAV to align itself with the slope, which complicates the task and increases the sensing needs. Ideally, the feet would respond optimally based on their position relative to the uphill and downhill directions, regardless of the drone's orientation. The landing strategy must carefully balance these competing needs to ensure energy dissipation and effective ice engagement by the spines. A detailed plan to manage these tradeoffs will be discussed in Section II.

C. CONTRIBUTIONS

The novelty of this research lies in the development of the specialized spiny foot for landing on ice, and in the study of its interaction with a constant-force suspension and RT. Specifically, the main contributions of this work are as follows.

1) An experimental investigation of ice–spine interaction under two scenarios: rapid vertical impacts (0.5–3 m/s) perpendicular to the surface, either through direct impacts or through a constant-force suspension, and low-speed horizontal sliding tangential to the surface.
2) The design of a novel passive spiny foot for landing on ice that utilizes the findings from the ice–spine interaction analysis to leverage the friction-based suspension and RT. The foot also allows the spines to engage late during landing and features two opposing retractable spines that generate anisotropic shear forces, while operating differently depending on their orientation with respect to the slope.
3) The experimental quantification and optimization of the system's touchdown envelope regarding allowable impact speed and ice slope angle, enabling landings on ice inclined up to 60° and at impact speeds of up to 3 m/s.
4) A field evaluation of the proposed solution through 24 landings on icebergs and glaciers in Iceland.

The remainder of this article is organized as follows. Section II outlines the global strategy and key system design principles for achieving safe landings on icebergs. Section III then presents the experimental study of ice–spine interactions, used to guide the design of the spiny foot. The complete system is then presented in Section IV, including the foot design, friction-based landing gear, UAV design, and landing state machine. The experimental touchdown envelopes are presented and discussed in Section V. Finally, Section VI describes the results and analysis of the field trials conducted in Iceland.

II. OBJECTIVES AND STRATEGY

The objectives of this study are to enable small multirotors to land on icebergs and glaciers of various shapes and sizes, remain landed for extended periods, and take off again to reposition or return to the launch site. Being able to land reliably on steep ice should provide a wide range of landing sites, making it easier to find a suitable place to land. In addition, sloped surfaces naturally drain water from melting, unlike concave regions that are easier to land on but more prone to flooding. This allows the drone to

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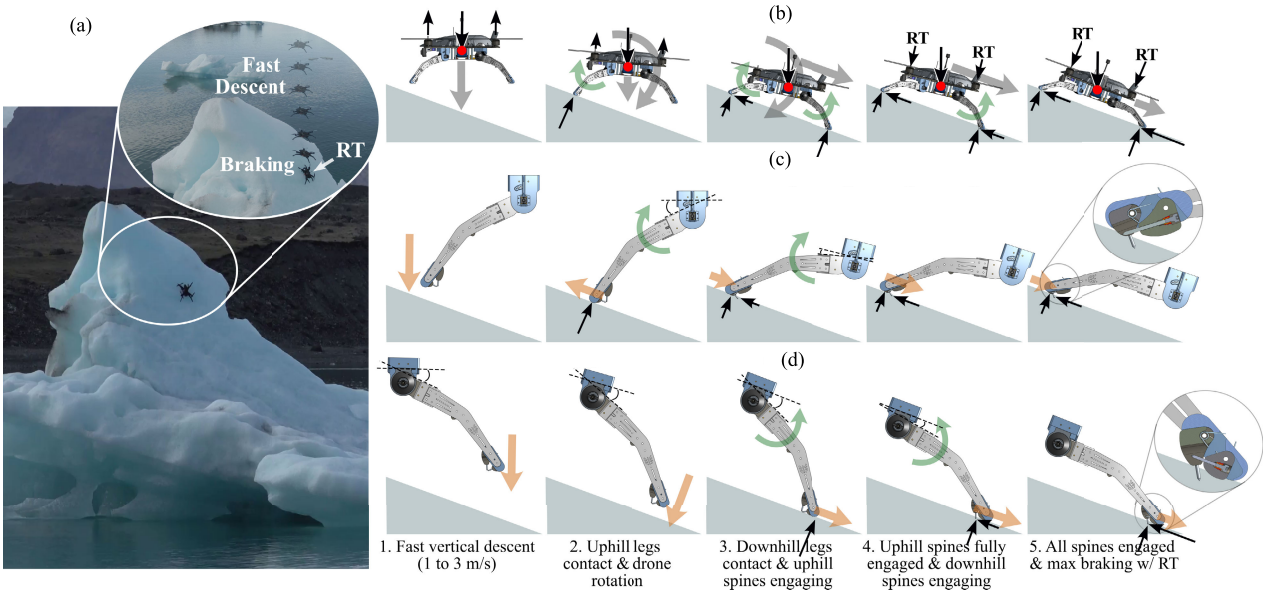


FIGURE 2. Illustration of the proposed strategy for landing on inclined ice surfaces. Gray arrows show the drone's movement; black arrows indicate the forces acting on the drone during landing; orange arrows illustrate the feet's movement; and green arrows depict the legs' rotation. Five frames demonstrating the different landing steps are shown and progress from left to right. (a) Expected landing sequence, where the drone descends at high speed (1–3 m/s), impacts the ice with a suspension to absorb the kinetic energy, and uses specialized spiny feet along with RT to aggressively stop its motion. (b) Side view of the drone during landing and impact, with the approximate location of the CoG represented as a red dot. (c) Detailed side view of the uphill legs during landing and impact. (d) Detailed side view of the downhill legs during landing and impact.

support longer stays without the risk of submersion. To have access to numerous landing sites, the drone must also be capable of landing on irregular and uneven ice surfaces caused by melting and calving. That being said, hazards such as holes or ice protrusions should still be avoided, which requires braking after touchdown within a short and predictable sliding distance. In the case of ice braking or melting under the spines, the feet should also be able to passively re-engage to prevent slipping. Finally, to operate near sea icebergs, the drone should be able to land in windy conditions of at least 30 km/h.

A. PROPOSED STRATEGY

To meet the listed design and mission objectives, the proposed landing strategy is divided into distinct phases, as shown in Fig. 2. The first step, once a landing site is selected, is for the drone to descend quickly at 1–3 m/s to minimize potential effects from disturbances until the uphill legs contact the ice slope. Such contact initiates Step 2, which causes compression of the uphill suspensions and outward rotation of these legs. This can make the feet move uphill, depending on the exact touchdown conditions. Note that it is undesirable for the spines to engage at this point, as doing so could generate large forces that push the drone downhill, prevent the suspension from absorbing the impact energy, or even break a leg. During this phase, the uphill spines do not engage the ice, both because of their anisotropic properties and because they require suspension compression to contact the surface. Once contact is made, RT is also initiated, which takes about 250 ms to reach its setpoint [24]. After initial contact, the drone's body rotates around the uphill feet in contact while also starting to move downhill, and the uphill legs continue to compress to dissipate energy. The uphill spines also gradually start to penetrate the ice. Step 3 begins when the downhill legs strike the ice. At this point, the uphill spines are almost fully engaged with the ice to slow the downward motion of the uphill feet, while the downhill feet can still slide freely, allowing further suspension compression and absorption of the remaining energy. In Step 4, the downhill suspensions compress, simultaneously lowering the drone's center of gravity (CoG) relative to the ground and engaging the downhill spines to start to generate braking forces. At this point, RT is also starting to take effect. This leads to Step 5, where all spines are fully engaged, the suspension is mostly compressed, and the RT is fully activated, allowing the spines to brake aggressively. The entire process is very short, limiting downslope travel to only a few centimeters. Once the drone is stable, RT is turned off to reduce energy consumption. It remains turned off until the next takeoff.

Since braking transfers the load to the downhill spines, these spines are sized larger to withstand these forces and prevent breakage. The uphill spines, although also involved in braking, have a smaller diameter to allow easy self-re-engagement under the minimal normal forces present on steep slopes. This is critical in the event of a slip occurring after a completed landing, when RT is deactivated. The downhill and uphill spines also vary in length and angles relative to the surface, chosen for optimal braking and passive re-engagement, respectively, based on the ice–spine

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interaction analyses covered in Section III. Finally, the spines are also sized to disengage through bending at a specific shear load rather than breaking, which can happen when they contact rocks or sediments trapped in the ice.

The rest of this section provides a detailed description of the strategy's core components.

1) FAST DESCENT

The first part of the strategy is to descend quickly toward the landing surface to minimize the time the drone is exposed to external disturbances [24], [25]. Indeed, strong wind gusts have been observed near large icebergs at sea [9]. By descending rapidly toward the landing site, the drone is less likely to be disturbed or deviate from its initial trajectory. This is especially important because drifting too far from the landing site could cause the drone to fall into a hole or crevasse or have its propellers hit ice protrusions.

2) IMPACT ABSORPTION WITH DISSIPATIVE SUSPENSION

The second key element of the strategy is to dissipate impact energy, which is crucial at higher descent speeds. This is accomplished with a constant-force dissipative landing gear, which reduces rebound to keep the spines engaged in the ice and limits peak forces, thereby lowering the risk of damage to the structure or electronics. This type of suspension has also been shown to be especially tolerant of variability in touchdown conditions, simplifying sensing, planning, and control [24]. Although attaching spines to the feet could restrict suspension movement and thus energy dissipation, their placement allows engagement only after significant compression, which is mainly caused by the impact dynamics. This enables both high energy dissipation and strong braking from the spines once sufficient compression of the legs is reached. This braking occurs when the drone's CoG is closer to the surface, and its stability is increased, further reducing the risk of flipping on steep slopes. Finally, the curved legs provide clearance, ensuring only the feet contact the icy surface, even on uneven terrain.

3) AGGRESSIVE BRAKING ON STEEP SLOPES WITH RT

To ensure the UAV stays close to the intended landing site, it is essential that the landing system allows the UAV to brake quickly within a minimal stopping distance. Failing to do so could cause the drone to fall into a crevasse or hit an ice protrusion with its propellers. To achieve the most abrupt braking, the downhill spine properties (i.e., diameter, length, and angle relative to the ice surface) are designed to maximize shear force, while also employing RT to boost the normal force on the spines, thereby increasing the shear force. The downhill spines are selected for the task of strong braking as they naturally experience higher normal forces from the drone's weight transfer during braking. To avoid breakage, the spines must be sized according to the loading conditions enhanced by RT. It is also important to note the critical role of RT during aggressive braking on steep slopes, where the normal force from gravity is substantially reduced (e.g., 50% of gravity on a 60° slope).

4) PASSIVE RE-ENGAGEMENT AFTER SLIP EVENTS

To prevent slips that occur after landing (e.g., ice melting and icebergs moving), when RT has been turned off, the spines must be able to re-engage quickly under low normal forces (i.e., steep ice slopes). Since the downhill spines are responsible for strong braking and are therefore larger, the uphill spines are tasked with the function of passive re-engagement. This is achieved by selecting properties of uphill spines (such as diameter, length, and angle) to produce the most shear force under minimal normal force. As shown in Sections III-D and III-E, this results in spines with a smaller diameter and length, making it easier to penetrate the ice. As a tradeoff, these smaller uphill spines generate less shear force than the downhill spines and have a smaller contribution during the aggressive braking phase. In addition, these spines must be sized to withstand high forces during the RT-augmented braking phase.

5) SPINY FOOT COMBINING BOTH FUNCTIONS

The specialized nature of the uphill and downhill spines typically requires aligning the feet with the slope before landing. However, both functions are integrated into a single foot design, as shown in Fig. 3. To accomplish this, each spine can rotate in opposite directions and is spring-loaded, so when one spine engages the ice during landing, the other spine passively retracts into the foot. By combining both spines within each foot, the system can function properly regardless of the drone's heading. This provides the UAV pilot with greater flexibility in how the drone is oriented during landing. That being said, to be clear throughout the rest of this article, the uphill spines will refer to those engaging and generating a braking force on the uphill legs, while the downhill spines will refer to those generating a braking force on the downhill legs.

III. ICE–SPINE INTERACTION

To properly size and design the spines for the feet, ice–spine interactions need to be investigated at a fundamental level to answer multiple questions. For example, how does momentum, kinetic energy, or work affect the penetration depth of a spine? Similarly, can a constant-force suspension favor spine engagement in the ice? What conditions lead to spine breakage and how can it be avoided? Can the spines be designed to release instead of breaking, as an overload protection? Unfortunately, the literature does not offer clear answers to these questions.

To answer these questions and inform the design of the foot and overall strategy, two experimental studies were performed. The first study involves perpendicular impact tests to investigate how factors such as impact speed, energy dissipation, and spine dimensions influence penetration depth. Results from these tests highlighted risks associated with ice landings using direct perpendicular impact, motivating

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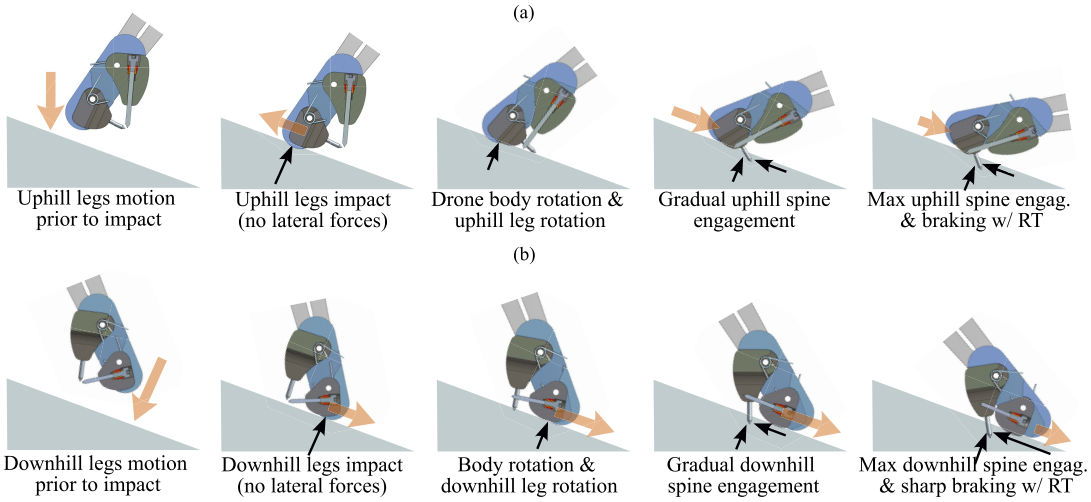


FIGURE 3. Integration of both the uphill and downhill spines into a single foot. Each spine is spring-loaded to retract itself when not used. Black arrows show the reaction forces, and orange arrows illustrate each foot's motion. (a) Detailed cross-sectional view of the uphill foot during impact. (b) Detailed cross-sectional view of the downhill foot during impact.

the second study, which consists of tangential traction tests to understand how spine properties and the forces acting on them influence shear force generation. The latter study helped define the global landing strategy discussed in Section II and determine the size of the spines.

Before proceeding to the two experimental studies, Sections III-A and III-B provide an overview of the ice-making process used to reproduce iceberg-like ice and the spines used in this work.

A. ICE-MAKING PROCESS

To study ice–spine–drone interactions, ice must be produced with representative properties. Glacier ice, especially from older and deeper layers, tends to have a dense and transparent-blue appearance because centuries of compression have removed air bubbles [26]. Denser, bubble-free ice is likely to make it harder for spines to penetrate and also increases shear resistance. The hardness of iceberg ice was measured through impact tests to be between 25 and 40 MPa, respectively, for temperatures between −0.5 °C and −14.5 °C [27]. The iceberg ice temperatures were also found to be fairly constant, even on warm spring days, ranging from −4 °C to 0 °C within the first 20 cm, consistent with other studies showing that the water–ice interface remains near 0 °C [28].

Studies on artificial ice in winter sports show that ice hardness varies with temperature according to [29]

$$P(T) = (-.6 \pm .4)\,T + (14.7 \pm 2.1) \tag{1}$$

where P is the hardness (MPa) and T is the temperature (°C). This relationship suggests that hardness values between approximately 12.6 and 26.8 MPa can be expected for ice between 0 °C and −10 °C. These values are lower than those reported for iceberg ice. However, it should be noted that the measurement methods varied, and that there is a lack of comparative studies enabling direct conversion between them. As such, these values should be interpreted as indicative of field variability rather than directly comparable quantities. The impacts and traction tests described in this section, using representative spine geometries, better reflect the forces generated during landing.

Transparent, air-free ice blocks were produced using directional freezing, a process where water gradually freezes from either the top or bottom, allowing expansion without cracking and reducing air bubble formation. Large open-top coolers filled with water were placed in a freezer from −12 °C to −15 °C for about 48 h, producing blocks roughly 8–12-cm thick. The ice–spine interaction tests were carried out outdoors in temperatures between −10 °C and −5 °C to achieve higher hardness and simulate worst case conditions. During warmer periods, the ice blocks were cycled in and out of the freezer, with surface temperature monitored by an infrared thermometer to keep it within −10 °C to −5 °C. Unless otherwise noted, this procedure was followed throughout the study.

It's important to note that ice making presents many challenges and that the results can be affected by temperature, air bubbles, crystalline structure, water purity, and other factors. As a result, the data obtained in Sections III-C and III-D exhibit variability that is difficult to eliminate fully. That being said, this variability is likely to also be found when landing on natural ice.

B. SPINES

The spines considered for this study are made of thin cylindrical rods of hardened M2 steel, manually sharpened to a 30° angle on a slowly rotating circular sandstone. M2 steel was chosen for its very high yield strength of about 1800–2000 MPa, high stiffness, very low plastic deformation, high hardness, and wear resistance [30]. This allows for the use of thinner spines, which eases penetration into the ice.

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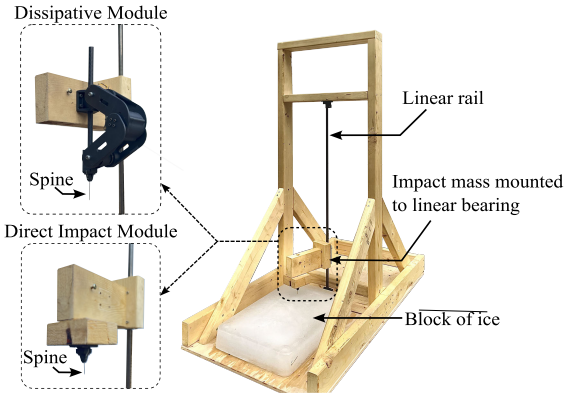


FIGURE 4. Test setup used to measure the penetration depth of a single spine in ice for varying impact speeds and masses, with and without a dissipative module. The two modules on the left are mounted on the linear rail. The rail has a height of 50 cm to reach impact speeds of 3 m/s.

C. PERPENDICULAR IMPACT TESTS

Penetration depth is a key element in determining the shear force a spine can produce. Studies have indicated that the penetration depth of an impactor during perpendicular collisions with ice either correlates with its momentum or its kinetic energy, depending on the impact regime, and is inversely proportional to ice hardness [29], [31], [32], [33]. However, for a spine mounted on a suspension, the penetration depth also depends on the suspension's force characteristics. Therefore, two tests were conducted to better understand the factors affecting penetration depth under typical impact conditions (i.e., impactor mass, speed, and presence of a force limiter) and spine sizes (i.e., diameter and length). The first test assesses penetration depth during direct impacts with the ice, while the second evaluates it when the spine is mounted on a constant-force suspension.

As shown in Fig. 4, the experimental setup used for both experiments consists of a vertical tower that drops a single spine mounted either directly under a mass or at the end of a constant-force damper. The spine is chosen to be long enough to prevent its plastic enclosure from reaching the ice surface during penetration, which causes rebounds. The dissipative module is similar to the dissipative landing gear on the drone, and is made of friction disks that produce a constant dissipative friction torque. For these experiments, the setup is configured to convert the spine's translational motion into a rotational motion of the dissipative module. For small angular motion, the resulting force on the spine is relatively constant. In both studies, penetration depth was analyzed based on momentum and kinetic energy, but it correlated better with the latter. To simplify results, only the kinetic energy graphs will be shown.

1) DIRECT IMPACTS

For the direct perpendicular impact tests, the following parameters were tested: spine diameters of 1.5, 2, and 2.5 mm, impact speeds of 1, 1.5, 2, 2.5, and 3 m/s, and masses of 915 and 1600 g. Multiple tests were conducted per combination to better account for ice variability, with ice temperatures ranging from −10 to −5 °C.

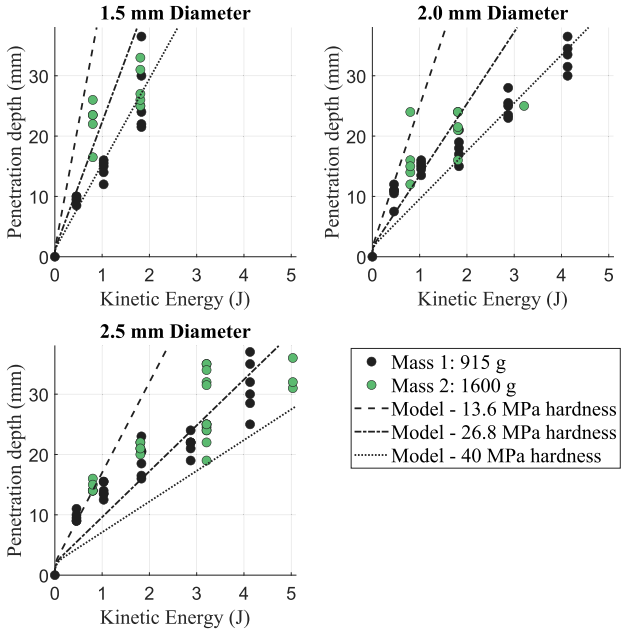


FIGURE 5. Penetration depth results for direct impacts and spine diameters of 1.5, 2, and 2.5 mm. The black lines represent modeled predictions based on kinetic energy at impact and displaced volume of ice for different ice hardness values [i.e., (4)]. Penetration depth was limited to 40 mm due to the spine's length.

Fig. 5 shows the penetration depth results, and slow-motion videos of the tests can be found in the online supplementary information. The penetration depths across all speeds and masses approximately form a cone, with the slope proportional to the kinetic energy and the diameter of the spine. Research on spherical indenters indicated that the impact kinetic energy K, the displaced ice volume V, and the ice hardness P are related according to [29]

$$V(d) = \frac{K}{P}. \tag{2}$$

In our case, the displaced volume of ice can be estimated as the volume of the cylindrical spine with conical tip in the ice, given by

$$V(d) = \begin{cases} \frac{1}{3}\pi d^3 \tan^2\left(\frac{\alpha}{2}\right), & \text{if } d < \frac{R}{\tan\left(\frac{\alpha}{2}\right)} \\ \pi R^2 d - \frac{2}{3}\pi \frac{R^3}{\tan\left(\frac{\alpha}{2}\right)}, & \text{if } d \geq \frac{R}{\tan\left(\frac{\alpha}{2}\right)} \end{cases} \tag{3}$$

where R is the spine radius, d is the penetration depth of the spine including the conical tip, and α is the spine's cone tip angle. When substituting (3) in (2) and rearranging, penetration depth d can be isolated and expressed as

$$d = \begin{cases} \left(\frac{3mv_i^2}{2\pi P \tan^2\left(\frac{\alpha}{2}\right)}\right)^{1/3}, & \text{if } d < \frac{R}{\tan\left(\frac{\alpha}{2}\right)} \\ \frac{mv_i^2}{2\pi PR^2} + \frac{2R}{3\tan\left(\frac{\alpha}{2}\right)}, & \text{if } d \geq \frac{R}{\tan\left(\frac{\alpha}{2}\right)} \end{cases} \tag{4}$$

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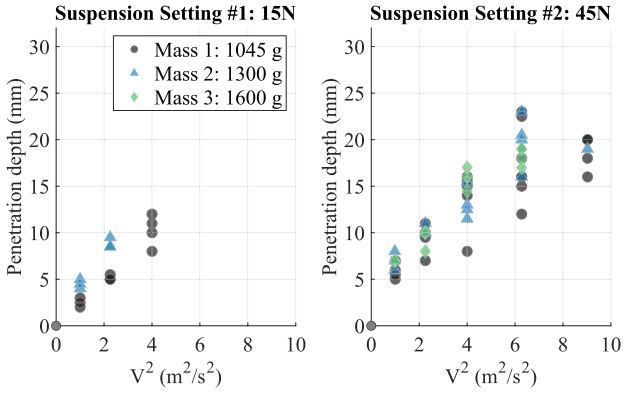


FIGURE 6. Spine penetration during impact tests for spines of 1.5-mm diameter, with a dissipative friction force of 15 ± 3 N (left) and 45 ± 5 N (right). Penetration depth is shown with respect to the squared velocity at impact. Both slopes are similar, suggesting that the penetration depth is related to the kinetic energy of the mass below the suspension (i.e., constant mass).

where v_i is the speed at impact. The predicted penetration depth is shown in Fig. 5, for ice hardness values of 13.6, 26.8, and 40 MPa. The first two values represent the expected hardness based on (1) for the temperature range used in the tests, and the last one represents the maximum hardness value expected from measurement on icebergs at −14.5 °C [27]. The experimental data observed in this test appear to match the prediction of (4) for penetration depths below approximately 15 mm, while deeper penetrations better fit the higher hardness level.

2) CONSTANT FORCE IMPACTS

Impact tests with the constant force suspension were also performed to understand the spine's penetration in these conditions. To minimize the number of tests, only a single spine diameter of 1.5 mm was tested. The following test parameters were used: masses of 1045, 1300, and 1600 g, impact speeds of 1–3 m/s in 0.5 m/s steps, and two friction settings of 15 ± 3 N and 45 ± 5 N (measured vertically at the spine).

These tests demonstrate that impacts occur in two phases. The first phase involves the direct impact of the lower part of the leg (i.e., spine, spine holder, part of the linkage, and the joint). This phase primarily stops the mass associated with these parts by penetrating the spine into the ice. Once the lower mass is stopped, the suspension starts to compress in the second phase, dissipating the energy associated with the upper mass. During this phase, the spine usually halts further penetration. If the impact energy is too high or the suspension friction is too low, a third phase occurs, involving impact between the upper and lower parts of the suspension. This causes additional spine penetration into the ice. However, as strong bottoming out is considered undesirable, it was avoided during testing.

Fig. 6 presents the penetration depth data for both suspension friction settings. The data were plotted with respect to the square of the impact velocity to show the relationship with the kinetic energy of the lower part of the leg. As expected, both suspension settings result in the same penetration trend. Assuming that the slope found for the direct impact tests (i.e., 22 mm/J) also holds in these tests, the slope found in Fig. 6 would suggest an effective mass of the lower part of the leg of approximately 250 g. For reference, the spine holder has a mass of 106 g while the joint's mass in this assembly is around 92 g. Slow-motion videos of a few impact tests conducted with the dissipative module (with both suspension friction settings) are available in the supplementary information found online.

In summary, both impact tests showed that penetration depth mainly depends on the kinetic energy of the mass that directly contacts the ice, i.e., below the dissipative joint when it is used. The dissipative joint thus decreases the penetration depth compared to direct impacts. For drone landings, deeper penetration of the spine is not necessarily desirable, as they can bottom out and rebound. To prevent this, longer spines are needed, but they are more prone to buckling, bending, and breaking upon impacts. This risk is even higher when landing on ice slopes, where the spines are likely to strike the ice at an angle. Deeply inserted spines could also get stuck and complicate takeoff. The depth achieved with direct impacts is also more sensitive to variations in impact speed, which demands more precise control. As will be seen in Section III-D, only a short spine penetration depth is needed to generate enough braking. Therefore, a dissipative suspension would allow a short, strong spine to more reliably penetrate the ice under varying touchdown speeds without leaving excess energy that could cause bounces and undermine spine engagement and shear force generation.

D. TANGENTIAL TRACTION TESTS

Following the initial impact and compression of the landing gear, the spines are being dragged through the ice to brake the drone. As seen in the field trials results (Section VI-B), the stopping distance could reach up to 25 cm. Tangential traction tests were conducted to better understand the behavior of spines dragged through ice under various loads and with different geometrical properties, such as spine diameter, free length, and spine angle. The results of these tests will help to adequately size spines to produce high shear forces for aggressive braking under high normal loads during the RT phase of landing, and to engage with the ice under low normal loads (no RT) at slopes of up to 60° in case of a postlanding slip.

To better visualize the variables of interest and the forces acting on a spine during a traction test, Fig. 7 shows a 2-D representation of the inclined spine, where θ is the angle of the spine relative to the ice plane, L is the spine's free length outside its plastic enclosure, and D is the spine diameter. It is assumed that the spine is free to translate in the x- and y-axes but is constrained from rotating about the z-axis, similar to the spines on the drone's feet. The traction force is applied

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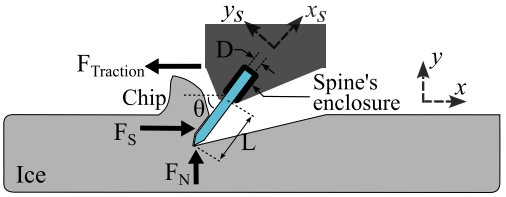


FIGURE 7. 2-D representation of a spine subject to tangential loading and machining of the ice surface.

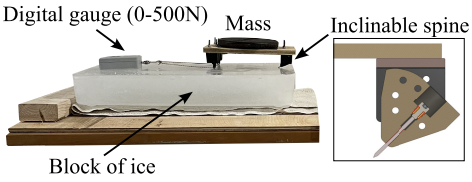


FIGURE 8. Test setup used to measure the resistive shear force of a single angled spine on ice.

as close to the ice as possible to minimize potential bending of the spine and assembly, which could cause early release.

The setup used for conducting the traction tests is shown in Fig. 8. It features a rigid plate that supports an adjustable mass, with 3-D-printed low-friction sliders on one side and a single spine on the other. The wide footprint between the low-friction sliders and the spine reduces rotation about the z axis and allows translations along the x- and y-axes. The spine is fixed at a specific angle using a 3-D-printed inclinable bracket with preset angles going from 30° to 75° in 15° increments. A threaded insert allows for the adjustment of its free length. The system is placed on top of a rectangular ice block and pulled over a distance of 10 cm, with a digital gauge measuring the peak force.

Three main operating modes were observed: ice machining, spine breakage, and spine skipping. During a traction test, the spine is expected to experience different operating modes depending on the exact loads and geometric parameters. The first mode, ice machining, occurs when the spine removes a layer of ice as it moves in the x-y plane, as shown in Fig. 7. Given that the spines used in this study are sharp, thin, and angled, even a small amount of normal load will cause the tip of the spine to engage the ice and machine a thin layer, compared to more traditional sliding between two surfaces.

The second mode, spine breaking, occurs when the shear force is such that the maximum stress in the spine exceeds its ultimate strength. Assuming the spine functions as a cantilevered beam with reaction forces at the tip, the maximum bending stress at the base of the spine can be expressed as

$$\sigma_{max} = \frac{32\ \mathrm{L}\,(F_N \cos(\theta) - F_S \sin(\theta))}{\pi D^3} \tag{5}$$

where F_N is the normal force (N) on the spine, and F_S is the shear force (N). Ideally, to achieve safe and reliable landings on ice, the spine parameters should be selected so that this mode is avoided.

The third mode, spine skipping, also observed in [5], occurs when the spine bends enough to escape the ice, before reengaging further down the ice. This scenario can be repeated multiple times as the spine continues to travel along the ice. Assuming again that the spine behaves as a cantilevered beam, the lateral displacement w of a cylindrical spine subject to the loads F_S and F_N applied at the spine tip can be written as

$$w = \frac{(F_S\ sin\,(\theta) - F_N\ cos\,(\theta))\,L^3}{3EI} \tag{6}$$

where E is Young's modulus and I is the second moment of inertia of the spine. Note that although the spine can be made as stiff as possible to prevent skipping, its ability to bend and skip under load was found to be beneficial. In fact, this behavior acts as a protection against spine breakage, which is valuable when landing in uncertain real-world conditions. Specifically, glacier and iceberg ice often contain small rocks that hinder normal ice machining by the spine, increasing forces and the risk of spine breakage.

To fully test the range of conditions applicable to a drone landing on ice, the following parameters were used: spine diameters of 1, 1.5, and 2 mm, free lengths of 4, 6, and 8 mm, normal forces on the spine of 1.6, 3.3, 6.3, 12.1, 17, and 23.1 N and spine angles of 30°–75° in 15° increments, for a total of 216 parameter combinations. As ice presents variability, traction tests were repeated five times for each combination of these parameters. The normal forces were chosen to consider the full range of normal loads experienced by the spines during braking, including the drone's weight, the slopes from 0° to 60°, and RT (more than two times the drone's weight).

Fig. 9 shows the experimental results of the traction test as a function of the normal force acting on the spine. Slow-motion videos of a few traction tests are also available in the supplementary information found online. The two columns represent spine diameters of 1.5–2 mm, and the rows represent the spine angles. Results for 1-mm spines are not shown because they broke at shear forces too low to be considered for the final spine foot design. The "x" markers indicate where breaking actually occurred. For comparison, dotted lines show the maximum theoretical shear force predicted by (5). The measured breaking points generally exceed this limit because it assumes the load is applied at the spine tip, while the actual resultant force acts somewhere between the tip and the base. Finally, two blue vertical dashed lines are added to each graph to indicate the normal load on a single foot during landing on a 60° slope, with and without RT.

Several trends can be observed from the test results. At low spine angles of 30°, the shear force is very minimal because the ice tends to fracture at the tip of the spine before penetrating deeper into the ice. At these low angles, the projected area of the spine on the ice, which generates the shear force, is very small. The forces also decrease at higher spine angles of 75° due to skipping. Specifically, at these angles, the spine needs less deformation to reach an angle that allows it to disengage. This is especially true for the

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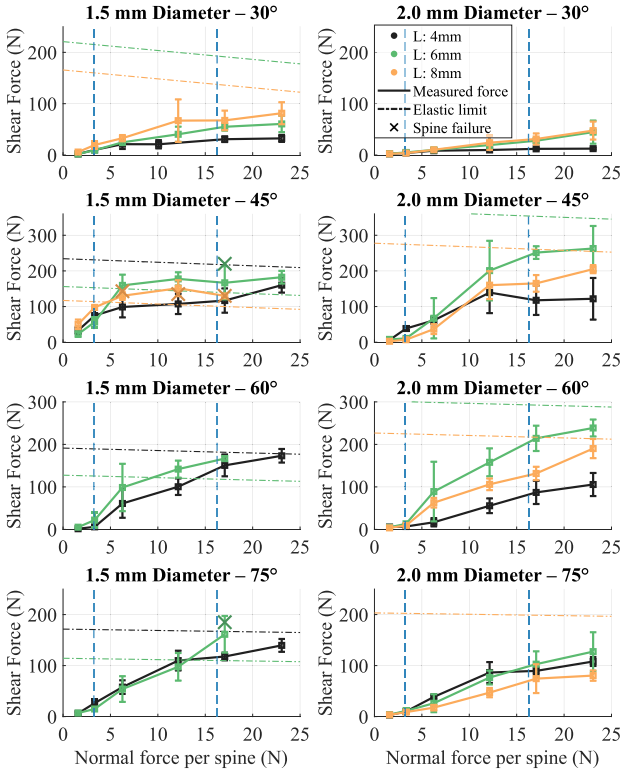


FIGURE 9. Traction test results showing the spine's average shear force and standard deviation as a function of the applied normal force, for spines of 1.5–2 mm in diameter. Blue vertical dashed lines show the normal load on a foot on a 60° slope with and without RT. The "x" markers represent a spine breaking during a test.

longer spines. For both high (75°) and low angles (30°), the experimental results show a relatively linear increase in shear force as the normal load increases.

At spine angles of 45°, and to a lesser extent at 60°, the shear force tends to converge toward a force plateau. This plateau defines the maximum force that the ice can sustain, which occurs when the spine's plastic enclosure contacts the ice, limiting the effect of further increases in normal loads. Ensuring that the shear force plateau remains well below the ultimate tensile strength of the spine allows the ice to fail before the spine. The 45° spine configuration offers an additional advantage: it produces high shear forces even at the lowest normal load level, which is crucial for the passive re-engagement during postlanding slip events. For example, the 1.5-mm spine inclined at 45° can generate between 55 and 95 N of shear force, depending on length, under a normal load of only 3.25 N. This means that the spine can generate up to 29 times the applied normal force in shear force. This is also well above the 5.6 N of shear force required for static equilibrium on a 60° slope.

Finally, at spine angles of 60°, the shear force plateau is less distinct than at 45° and shows a more linear increase, while the spines generate lower shear forces at low normal loads. Some skipping also occurs with the longer spines, which explains the lower shear force recorded on the 8-mm-long spine compared to the 6 mm ones. Although the 60° spines produce less shear force than the 45° spines at very low normal forces (under 5 N), their overall behavior makes this angle suitable for aggressive braking under RT. Indeed, they exhibit a steadier increase and do not reach a clear plateau at the normal forces of interest, unlike the 45° spines. This indicates that RT can be used to achieve very high braking forces. For example, a drone in static equilibrium on a 60° slope, applying RT equal to twice the drone's weight, would increase the normal force on the ice by a factor of five compared to gravity alone. This would amplify the shear force by approximately 15.6 times for the 1.5-mm spine diameter, averaged across all spine lengths. This increase in shear force highlights the critical role of RT in landing on steep ice. Finally, the skipping observed at spine angles of 60° acts as an extra protection during real-world tests on ice surfaces with rocks or sediment, serving as a safeguard against unexpected surface conditions not present in laboratory tests.

E. SPINE SIZE AND ANGLE SELECTION

Using the tangential traction test results, the geometric parameters for the uphill and downhill spines can be chosen to fulfill the two main functions: aggressive braking under high normal loads during thrust reversal and spine re-engagement under low normal loads in case of postlanding slip events. Note that the design details on how the two spines were integrated into a single foot to produce anisotropic shear forces will be discussed in Section IV-A.

Since weight is transferred to the downhill legs of the drone during braking, the downhill spines were chosen for their ability to handle aggressive braking and generate high shear forces. Selecting the size and angle of the downhill spines involves balancing two criteria: maximizing shear force at high normal forces of 16.25 N (i.e., twice the drone's weight in RT on 60° slopes), while also allowing for skipping when needed to provide additional safety against unexpected ice conditions. To meet these criteria, the following parameters were selected: spine angle of 60°, diameter of 2 mm, and length of 8 mm. Although the 6-mm spines can generate more shear force, the longer spines can limit shear force by skipping out, which is especially useful when contacting rocks in the ice. This acts as an overload protection for the spine.

Sizing of the uphill spines, responsible for re-engagement under low normal loads, involves once again balancing two criteria: maximizing shear force at a normal force of 3.25 N while keeping the shear force plateau safely below the spine's ultimate tensile strength. The latter criterion is only important during the RT phase, where the high normal forces could risk breaking the uphill spines if the shear force plateau meets the spine's tensile limit. The final chosen parameters are a spine angle of 45°, a diameter of 1.75 mm, and a length of 6 mm. Interpolating between the 1.5 and 20-mm results, the 1.75-mm diameter was chosen to avoid

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spine breakage while featuring a sharp increase in shear force at low normal forces and a distinct shear force plateau.

IV. SYSTEM OVERVIEW

This section provides a comprehensive overview of the systems designed for landings on steeply inclined ice. It covers the design of the spiny foot, the friction-based dissipative landing gear, and the custom UAV platform known as the *Ice Dart*. Fig. 10 illustrates the hardware used. In addition, it discusses the software implementation of the landing state machine, which activates RT upon impact.

A. SPINY FOOT DESIGN

In Section II-A, a strategy was proposed to enable a single foot to perform both strong braking and easy self-re-engagement by passively engaging either of two spines based on the sliding direction. In Section III, experimental studies of ice–spine interaction were used to determine the diameter, length, and angle of the spines for both functions. In this section, these findings are combined into a final foot design. The spiny foot design is shown in a cross-sectional view in Fig. 10 (right), and the actual foot is shown in Fig. 11. The foot consists of two spring-loaded, rotating spine systems configured to rotate in opposite directions. This setup allows one spine to retract into the foot while the other engages with the ice, enabling each foot to passively engage the appropriate spine regardless of how the drone is oriented relative to the angled ice, as shown in Fig. 3.

Each spine-rotating assembly consists of a single spine that is press-fit into a 3-D-printed rotating plastic support. It includes an insert and a screw, allowing the spine to rest on a hard surface and enabling easy length adjustments if necessary. The angle of the spines in the foot is determined by the final orientation of the legs after the suspension has fully compressed. This design ensures that the spines do not engage into the surface upon first contact. It also permits the legs to rotate freely during the initial impact phase to dissipate energy and protects the spines from high-impact forces. As the suspension compresses, the spines gradually penetrate the ice, reaching their final position and maximum shear force when the suspension is fully compressed.

B. FSA LANDING GEAR

The friction shock-absorbing landing gear, shown in Fig. 10 (left), represents a new implementation of a similar design documented in previous work for landing on rooftops [23] and high-speed vehicles [24]. To summarize the main features of the previous landing gear design, the FSAs are comprised of 38 friction disks that produce friction torque as the leg rotates upon impact. A highly geared 6-V DC motor and leadscrew control the pressure on the disks to adjust the friction torque before landing. After each landing, the pressure is released, allowing gravity and small internal torsion springs to reset the legs for the next landing. This FSA design was originally scaled to absorb the energy of a 2.4-kg drone landing at 4 m/s with a safety factor of 1.2. For the 2.65-kg *Ice Dart*, the corresponding maximum landing speed would then be 3.8 m/s. Although the *Ice Dart* is intended to land at speeds below 2 m/s on icebergs and glaciers, studies indicate that strong wind gusts could cause significant speed variations [9]. This prompted us to keep the current FSA design for an extra safety factor, even though it reduces flight time. In addition, it was found that a considerable amount of energy is dissipated by the spines penetrating and breaking the ice, compared to rubber feet sliding over a hard surface during landing.

C. ICE DART UAV

The *Ice Dart* UAV depicted in Fig. 10 (center) was designed to provide a high-power, high-maneuverability quadrotor platform that can fly stably in strong winds, withstand high impact speeds of up to 3 m/s, and reach icebergs at distances of more than 500 m while transmitting stable telemetry and video data.

To achieve the high power and maneuverability needed to navigate the potentially strong winds near large ice structures, the propulsion system features 1050 KV motors (Brotherhobby Tornado T5), $10 \times 5 \times 2$ in propellers, and a 4-in-1 65 A ESC. This configuration results in a thrust-to-weight ratio of 6. The entire system is powered by a 6S 4 Ah 100C LiPo battery, providing a flight time of approximately 10 min.

To withstand high-impact speeds, the frame is made of custom, CNC-machined carbon fiber parts. The drone has a span of 45 cm from motor to motor (diagonal) and weighs a total of 2.65 kg, including the 400-g FSA landing gear.

The drone's flight controller is a Cube Orange Plus from CubePilot running the Ardupilot firmware. To ensure stable flight in strong winds, the *Ice Dart* is equipped with the F9P RTK GNSS module, a multiband, multiconstellation GNSS receiver that provides centimeter-level accuracy when paired with an RTK base station. The RTK feature was ultimately used in some field trials, but not all, as the base antenna broke during one of the trials. An RM3100 compass was also added to improve yaw measurement accuracy.

For reliable communication with the ground control station (GCS) when landing on icebergs at 500 + m, the drone is equipped with a SIYI HM30 long-range digital transmission system, which manages all communications between the drone, GCS, and pilot, including the telemetry, visual feed from the onboard camera, remote controller commands, and GNSS-RTK corrections. The transmission range was validated at up to 5 km in direct line of sight.

To ensure the drone could float and be recovered if it falls into the water, a custom closed-cell EVA foam shell was shaped to fit the drone, with additional foam layers added underneath the body and between the carbon-fiber plates of the legs. Based on conservative volume estimates from the computer-modeled UAV, the 1964 cm^3 (57 g) of foam should enable the drone to float with a 15% safety margin. The top and bottom foam shells also provide the secondary benefit of shielding the UAV electronics from light rain or

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snow, which can allow for longer testing periods in harsh weather conditions.

A free-body diagram of the *Ice Dart* on an inclined surface, shown in Fig. 12, illustrates the influence of the CoG on stability margins. The drone remains in static equilibrium as long as the CoG lies within the horizontal area formed by the uphill and downhill foot contact points ($c \geq 0$). This condition results in a maximum slope angle of approximately 71° before tipping occurs, assuming the downhill spines generate sufficient shear force to act as a pivot point. For comparison, a DJI F450 drone equipped with similar hardware (battery, frame, propulsion system, etc.) and its standard rigid landing gear has a CoG height of approximately 133 mm and a lateral distance of 193 mm from its feet, resulting in a tipping limit of about 35°. The *Ice Dart*'s large increase in stability margin reduces its likelihood of flipping and instead favors sliding, which is subsequently mitigated by the high shear forces generated by the spines.

Finally, to improve robustness to heading offsets relative to the slope gradient, which inherently lead to asymmetric foot contact at impact, the *Ice Dart* was designed with a 30° outward leg stance. This configuration provides two key benefits. First, it ensures that at least two legs can effectively engage their spines for a large range of headings. In comparison, a specialized configuration with parallel legs would be highly performant with perfect headings, but all legs lose performance with heading offsets. Since a single spine can generate large braking forces when fully engaged, it is considered preferable to have one pair of top and bottom legs engaging effectively (i.e., an "X" configuration) rather than four legs partially engaging. Second, the outward leg stance, compared to a parallel legs configuration, increases the vehicle's lateral footprint, thereby improving stability about the roll axis. This becomes increasingly important for landings with large heading offsets. Overall, this configuration should reduce the need for precise alignment of the heading with the slope, thereby alleviating pilot workload.

D. SOFTWARE INTEGRATION

The UAV follows a simple landing sequence programmed in Ardupilot and used in related work for landing on rooftops [23]. When the operator switches to this custom landing mode, the UAV descends at a fixed, constant velocity until it detects contact using its onboard IMU. Contact detection is based on a threshold applied to the vertical acceleration estimate provided by the attitude heading reference system (AHRS) in ArduPilot. The threshold is set to -16 m/s^2, which includes gravity measurements. For reference, flight data logged during the field trials show that the measured acceleration remains between -6 and -10.5 m/s^2 during descent, whereas the impacts produce peaks of at least -25 m/s^2. The selected threshold, therefore, provides a large margin between in-flight and impact conditions. Overall, no incorrect contact detections were observed across more than 50 landings during trials, at impact speeds ranging from 0.5 to 3 m/s (including experimental tests and preparatory trials).

After the first contact detection, RT is activated and maintained until the drone is stably landed, or after 6 s have passed. The RT magnitude was selected based on a static force balance to ensure sufficient normal force on the downhill feet for effective spine engagement and aggressive braking. Traction tests (Fig. 9) indicate that approximately 11 N of normal force per downhill spine is needed to generate shear forces on the order of 100 N for the selected spine geometry. As the normal force contribution from gravity decreases with increasing slope angle, RT varies to maintain this threshold. Based on static equilibrium calculations and on the propulsion system's thrust capacity, an RT level of 60% is used on slopes up to 30°, 80% on slopes between 30 and 50°, and 90% above. This strategy provides 1.5–5 N of additional normal force per downhill spine, for safety. Note that the landing site slope angle is estimated prior to landing during field trials, and the RT level is preset accordingly. During impact, RT is also automatically increased to 90% (about 42 N of total force) if the drone detects that it is starting to flip.

Finally, the autopilot was also modified to ensure the drone always returns to the ground station in the event of a failsafe, rather than returning to the previous takeoff location (e.g., return-to-launch). Indeed, the launch site could be an iceberg or glacier if the failsafe occurs after these landings.

V. LANDING ENVELOPES

To understand how the proposed spiny feet, dissipative landing gear, and RT enable landing on steep ice and passive re-engagement, two experiments were conducted in this section. First, to determine under which conditions the *Ice Dart* can land safely, the experimental touchdown envelopes were generated based on the impact conditions of interest (i.e., ice surface inclination and touchdown speed). Specifically, the individual contributions of the uphill spines, downhill spines, and RT were examined. Then, slip tests were performed to assess the drone's ability to re-engage its spines after a slip incident, comparing the uphill and downhill spines separately.

To conduct these tests, an inclinable ice platform was built onto which the drone can be dropped at different impact velocities, as shown in Fig. 13. The ice surface, supported by a 0.9×1.2-m wooden platform, can be inclined from 0° to 60° in 15° increments. The impact velocity can be set by manually adjusting the drone's drop height. The drone is released via a quick-release latch mechanism from Sweeney Special Effects. An infrared thermometer is used to monitor the ice surface temperature. All the tests were conducted on hard, transparent ice at temperatures ranging from -5 to -10 °C.

To quantify the individual contributions of the uphill spines, downhill spines, and RT, four different configurations were tested: uphill spines only without RT, downhill spines only without RT, both uphill and downhill spines without

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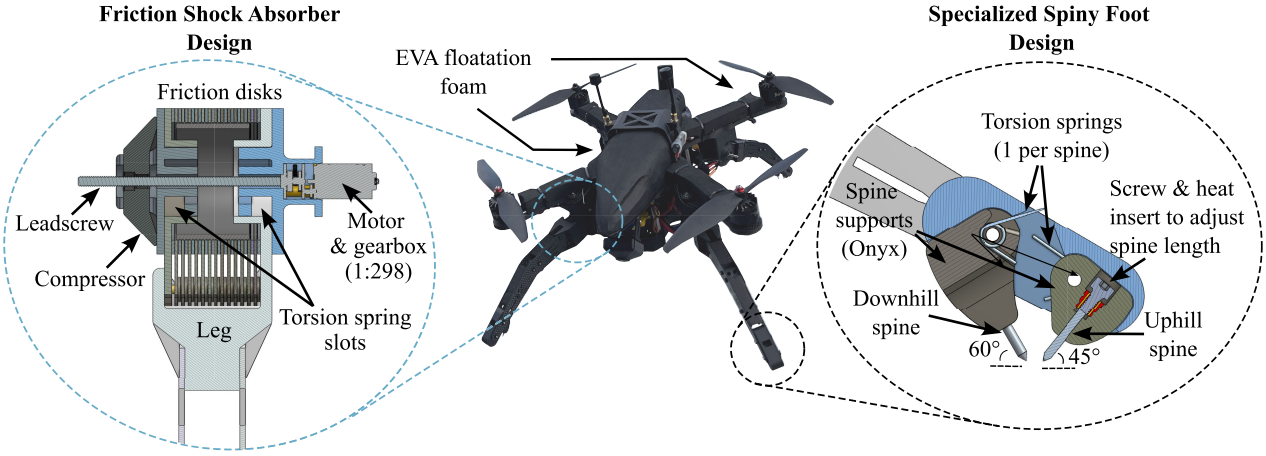


FIGURE 10. **Hardware overview, including a detailed view of the friction shock-absorbing landing gear and the novel specialized spiny foot design. The final design has an uphill spine of 1.75-mm diameter and 6 mm of free length, and a downhill spine of 2-mm diameter and 8 mm of free length.**

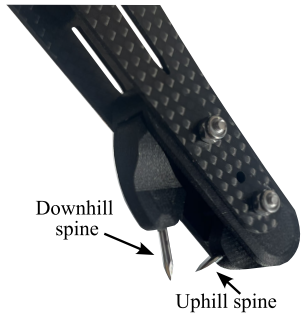


FIGURE 11. **Close-up view of the final spiny foot design.**

RT, and both uphill and downhill spines with 75% RT. For the touchdown envelope generation, a fixed RT value of 75% was chosen as it is theoretically sufficient to generate enough normal forces for the spines to provide large shear forces for the entire range of slope angles tested, at up to 60°. For configurations using only one set of spines, the other set was completely removed from the feet. The FSA landing gear settings remained unchanged throughout all tests. For each of the four configurations, ice surface angles of 15°, 30°, 45°, and 60° and impact speeds of 0, 1, 2, and 3 m/s were tested. Impact speeds were estimated from the drop height measured at the uphill feet.

The touchdown envelopes obtained for all four configurations are shown in Fig. 14, and a few videos of the drop tests are available in the supplementary material found online. The first point to note is that the uphill spines by themselves (top left) performed well at impact speeds below 2 m/s on ice inclinations of up to 30°. However, they provided insufficient braking at higher speeds and steeper slopes. Conversely, the touchdown envelope of the downhill spines alone (top right) shows a significant improvement in landing performance at higher impact speeds and steeper slope angles. This improvement can be attributed to the specialized design of the downhill spines for more abrupt braking.

Interestingly, none of these drop tests caused the drone to flip, even on 60° slopes, at high impact speeds, and without RT. This can be explained by three factors: the suspension absorbs the impact and prevents rebounds; the drone's CoG lowers as the suspension compresses; and the peak braking forces occur only once the suspension is fully compressed, avoiding large torques that could cause flipping. This behavior suggests that the drone could likely stop itself if given a sufficiently long braking distance. However, a 20-cm limit was set to minimize the risk of sliding into hazards such as crevasses or ice protrusions. These two touchdown envelopes also show that the downhill spines allow the drone to land in about 33% more conditions than when using the uphill spines.

Only a slight increase in performance is observed when using both the uphill and downhill spines, as seen in Fig. 14 (bottom-left). This confirms that the downhill spines do most of the braking, given their specialized design and because they naturally experience additional weight transfer during braking.

It can also be seen in Fig. 14 that the performance gain from adding RT is substantial, covering 90% of the landing conditions included in slope angles up to 60° and landing speeds up to 3 m/s. This is because the normal forces from gravity alone on a 60° slope are extremely limited (i.e., 3.25 N per foot, on average, in static equilibrium). These increase to about 11–12 N per foot with 75% RT, which allows the spine to generate very large shear forces for aggressive braking.

One piece of information that is not captured in the landing envelope is the spine engagement length. For landings with RT, near-complete spine engagement is observed, as increased normal forces promote deeper spine penetration into the ice. In scenarios without RT, however, partial spine engagement is more likely. This is caused by minor rebounds, spine skipping, or if the *Ice Dart* reaches static equilibrium before full penetration is achieved. That being

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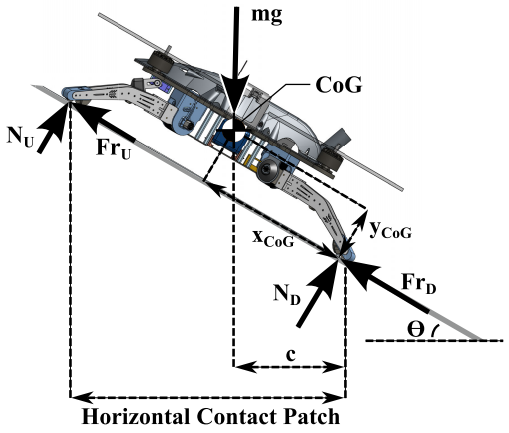


FIGURE 12. Free-body diagram of the *Ice Dart* on an inclined surface. The CoG is located at distances x_{CoG} and y_{CoG} of 264 and 93 mm, respectively, relative to the feet contact points in the fully compressed leg configuration. Subscripts U and D denote the uphill and downhill legs, respectively.

said, even for relatively short penetration depths (e.g., 4-mm-long spines), substantial shear force can be generated for spine angles between 45° and 75°, as presented in Fig. 9. Partial spine engagement is unlikely in the field and does not constitute a critical failure mode, as RT was consistently employed during field trials.

The final validation aimed at validating the uphill spines' ability to passively re-engage with the ice under low normal loads, as would be the case for a spine that would slip postlanding without RT. To quantify that ability, the drone was manually placed on the inclined ice with its suspension already compressed and the spines slightly above the surface, then released. To isolate the contribution of each set of spines, only one set (uphill or downhill) was tested at a time while the other set was removed, keeping the feet in contact with the ice. The process was repeated three to six times for each spine configuration and platform angle, for angles of 30°, 45°, and 60°. At 30° and 45°, both the uphill and downhill spines could stop the drone within 20 cm. At 60°, however, the uphill spines re-engaged with the ice in 66% of the slide tests, while the downhill spines only re-engaged 17% of the time. This outcome was expected, as the uphill spines were specifically designed for easy re-engagement.

VI. FIELD TRIALS

Field trials were carried out to validate the effectiveness of the proposed solution for landing on icebergs and glaciers. Specifically, the trials aimed to confirm that the proposed solution enables landing on steep, irregular ice surfaces of up to 60° at high impact speeds, across various ice types and impurity levels. The field trials also intended to demonstrate that the rapid landing approach remains reliable in strong wind conditions. Southern Iceland was chosen for its year-round access to coastal icebergs and glaciers, enabling land-based testing without the added complexity of operating from a boat. The trials took place in late April and early May, when icebergs are typically more abundant, daylight is longer, and weather conditions tend to be more favorable (i.e., windy but with less precipitation). All tests were conducted at the following three locations.

1) *Site 1:* Lagoon at the end of Sólheimajökull glacier.[1] This site was chosen for the proximity of the icebergs to the shore to practice the operations.
2) *Site 2:* Jökulsárlón, a lagoon at the end of the Breidamerkurjökull outlet glacier.[2] This site was selected for its abundance of large icebergs with steep surfaces at various distances from the shore.
3) *Site 3:* Fjallsjökull glacier across the Fjallsárlón lagoon.[3] This site was chosen to demonstrate landings on a glacier.

The rest of this section describes how the operations were conducted and documents the results obtained during the trials, along with key lessons learned in the field.

A. MISSION AND OPERATIONS

The mission was divided into five phases: scouting, landing site selection, alignment, landing, and return to launch. The complete mission profile is shown in Fig. 15. During the scouting phase, potential icebergs or glaciers were identified remotely from the GCS using a monocular rangefinder (6× zoom, 1200-m range) to ensure landing sites were within 600 m. This range limit was set so that enough time was available for the full mission. During the landing site selection (Phase 2), a second drone, the DJI Mavic 2, was used by the team to visually inspect the iceberg or glacier and identify suitable landing zones of at least 1 × 1 m, inclined at no more than 60°, and free of major holes or protrusions. These broad criteria greatly facilitated rapid identification of suitable sites. Since this study focuses on pushing drone landing performance rather than automated site selection, this dual-drone approach was adopted to reduce operational risks and ensure precise alignment without relying solely on the *Ice Dart*'s onboard camera. Once a site was selected, the *Ice Dart* was flown to the same coordinates at a lower altitude and manually aligned above the site (Phase 3) through coordinated communication between the two operators. The Mavic could then be repositioned to film the landing, and the automatic landing sequence described in Section IV-D was activated (Phase 4). After touchdown, the *Ice Dart* performed a normal takeoff, and both drones returned to the launch point (Phase 5). Because most landing targets were small icebergs or bergy bits prone to rolling, the *Ice Dart* remained landed on the ice with the propellers off only briefly, typically a few seconds to a few minutes.

All the equipment, including the drone, batteries, propellers, antennas, tripod-mounted radio system, tripod-mounted GNSS-RTK system, and GCS laptop fit into a single Pelican case and can be deployed within 10–15 min.

[1]GCS location: 63.535224°N, –19.349658°W.
[2]GCS locations: 64.061815°N, –16.170180°W; 64.057958°N, –16.175522°W; 64.045481°N, –16.219397°W; 64.044488°N, –16.201538°W.
[3]GCS location: 64.012496°N, –16.414645°W.

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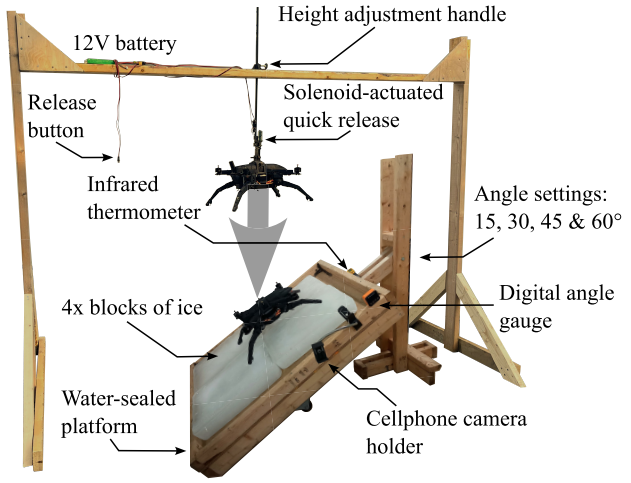


FIGURE 13. Drop test rig used to generate the touchdown envelopes experimentally. The drone is suspended at a specific height to achieve different touchdown speeds.

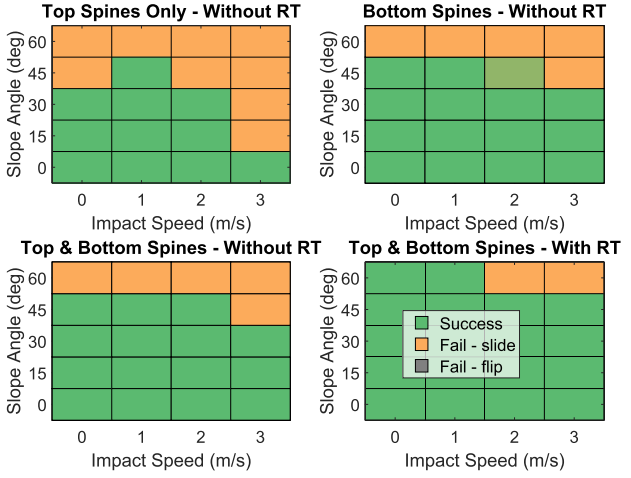


FIGURE 14. Experimental touchdown envelopes for different impact speeds, slope angles, and spine configurations, with and without RT. Failure from sliding is considered when the drone slides after the first contact for more than 20 cm. Failure from flipping was not observed.

Fig. 16 shows the GCS setup deployed at the third test site. The GCS laptop monitors the *Ice Dart*'s status before and during the mission, while a secondary screen next to it displays the onboard camera feed. The two drone operators are close to each other to ensure clear communication during the mission.

B. FIELD TRIALS SUMMARY AND RESULTS

A total of 24 landings were carried out over three days, all of which were successful. This includes two landings on glaciers and 22 on icebergs. Overall, the drone landed on 14 icebergs and two different glacier faces. Multiple landings on the same iceberg, but at different sites, were also performed to showcase the extensive landing options provided by the proposed solution. Fig. 17 shows all the landing sites where the *Ice Dart* landed during the field trials, and Table 1 provides additional details about each landing, including ice conditions, landing speed, RT levels, and final slope angle. Together, the figure and table illustrate the variety of iceberg sizes, surfaces, ice types, and landing conditions tested. The landings are numbered sequentially, with the difficulty generally increasing over time, as easier landings were performed initially to help the team become more familiar with the operations. In addition, footage from the field trials is available in the supplementary information found online.

Based on the Manual of Standard Procedures for Observing and Reporting Ice Conditions (MANICE) [34], landings were conducted on nearly all types of iceberg shapes: tabular, nontabular, domed, pinnacle, and wedge. As shown in Table 1, the *Ice Dart* also landed on three iceberg size categories: 6 bergy bits (i.e., height of 1–5 m above water), ten small icebergs (i.e., height of 5–15 m above water), and six medium icebergs (i.e., height of 16–45 m above water). Iceberg heights were estimated using the *Ice Dart*'s logged GPS altitude at its various landing sites.

A wide range of ice conditions was encountered during the trials, from very dirty ice (i.e., a mix of sediments, rocks, and ash that accumulated from the prolonged melting of dirty layers) to clean white or pale-blue ice. An additional mosaic of close-up images from the *Ice Dart*'s onboard camera is also available in the Supplementary Material, showing the ice surfaces in more detail. Blue ice indicates fresh, clear, and bubble-free ice with minimal fractures, while white ice is typically covered in micro-fractures caused by thermal shock (e.g., rapid heating after an iceberg flips and exposes a new surface to the sun). Tests 21 and 22, including the steepest landing at 58° (test 22), were conducted on clear, blue ice. Tests 1–8 at Sólheimajökull were on dirty, rock-filled gray ice, while the remaining landings, including those on glaciers, took place on mostly white ice with minor impurities. It is worth noting that, despite landing on a variety of ice conditions and impurity levels, the same set of spines was used throughout all trials without failure. In addition, no wear, tear, or dulling of the tips was observed. However, a slight reduction in the smooth rotation of the spines' plastic enclosures was observed due to dust and dirt ingress.

As shown in Table 1, the *Ice Dart* landed at impact speeds between 0.99 and 1.55 m/s, on slopes up to 58°, and in winds up to 30 km/h as measured at the GCS. By default, the target descent speed and RT were set to 1.5 m/s and 60%, respectively. However, when landing was expected to be near the touchdown envelope boundaries, descent speed was reduced to 1 m/s, and RT was increased up to 90%. The descent and impact speeds generally remained very close to the target descent speed, with a standard deviation of 0.032 m/s relative to the target, averaged across all tests. Landings were conducted on icebergs at distances up to 554 m from the GCS, with an average of 220 m.

It is also interesting to note that the drone had a minor slip incident during test 22, where its uphill right foot slipped by a few millimeters after landing and after RT was dis-

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TABLE 1. Compilation of the field trial results.

Test ID	Iceberg ID	Iceberg size class[a]	Ice type	Descent Speed[b] (m/s)	RT (%)	RTK used	Distance from GCS (m)	Descent length (m)	Descent Duration (s)	Recorded Wind[c] (km/h)	Impact speed (m/s)	Heading Offset[d] (°)	Landed Angle (°)
1	1	Bergy Bit	Dirty, rough	1.5	60	RTK	37	2.8	2.34	9-20	1.46	N/A	8
2	1	Bergy Bit	Dirty, rough	1.5	60	RTK	37	2.7	2.21	9-20	1.44	N/A	2
3	2	Bergy Bit	Dirty, rough	1.5	60	RTK	24	2.4	2.02	9-20	1.51	32	15
4	2	Bergy Bit	Dirty, rough	1.5	60	RTK	23	2.8	2.26	9-20	1.48	N/A	13
5	2	Bergy Bit	Dirty, rough	1.5	60	RTK	22	2.3	1.85	9-20	1.51	9	24
6	2	Bergy Bit	Dirty, rough	1.5	60	RTK	22	2.6	1.93	9-20	1.49	9	29
7	3	Small	Dirty, rough	1.5	60	RTK	245	3.1	2.26	9-20	1.50	26	35
8	3	Small	Dirty, rough	1.5	60	RTK	244	3.4	2.54	9-20	1.49	26	36
9	4	Small	Clean, clear, white, rough	1.5	60	No RTK	85	2.6	2.02	20-30	1.44	10	25
10	5	Small	Slightly dirty	1.5	60	No RTK	119	1.8	1.59	20-30	1.50	N/A	6
11	6	Small	Clean, clear, white, rough	1.5	60	No RTK	169	4.3	3.17		1.55	22	33
12	7	Small	Clean, clear, white, rough	1.5	60	No RTK	217	2.7	2.18	20-30	1.45	1	20
13	8	Small	Clean, clear, white, rough	1.5	60	No RTK	169	4.7	3.47	20-30	1.48	6	33
14	9	Medium	Clean, clear, white, rough	1.5	60	No RTK	554	6.2	4.43	20-30	1.48	21	18
15	10	Small	Clean, clear, white, rough	1.5	60	No RTK	328	4.6	3.60	20-30	1.47	6	23
16	11	Small	Clean, clear, white, rough	1.5	60	No RTK	242	9.0	6.30	20-30	1.48	N/A	10
17	12	Small	Slightly dirty	1.5	60	No RTK	412	5.1	3.85	20-30	1.43	0	23
18	13	Medium	Slightly dirty, rough	1.5	60	No RTK	367	3.5	2.61	7-8	1.48	34	17
19	13	Medium	Slightly dirty, rough	1	80	RTK	357	8.0	8.18	7-8	1.00	2	46
20	13	Medium	Slightly dirty, rough	1	80	RTK	382	3.2	3.36	7-8	0.99	4	32
21	14	Medium	Clear, blue, smooth	1	80	RTK	164	5.4	5.27	7-8	1.00	32	48
22	14	Medium	Clear, blue, smooth	1	90	RTK	168	4.8	4.48	7-8	0.99	3	58
23	15	Glacier	Clean, white, rough	1.5	80	RTK	424	5.1	3.80	7-8	1.46	45	42
24	15	Glacier	Clean, white, rough	1.5	80	RTK	446	6.8	4.92	7-8	1.51	31	32

[a] According to the MANICE classification system [34].

[b] Target descent speed, set as a parameter in Ardupilot.

[c] This column shows the wind speed range measured with an anemometer at the GCS during each day.

[d] Heading offset relative to the slope. Marked as N/A whenever the slope angle is less than 15°, as it is estimated with the IMU and requires sufficient surface inclination.

abled. Fortunately, the uphill spines reengaged immediately. Interestingly, substantially less sliding was observed on ice than on rooftops because the engagement of the spines in ice can stop the drone much faster than rubber feet on roof shingles. Finally, as observed in Fig. 17, for most landings, the drone ended up close to the ice with its body low to the

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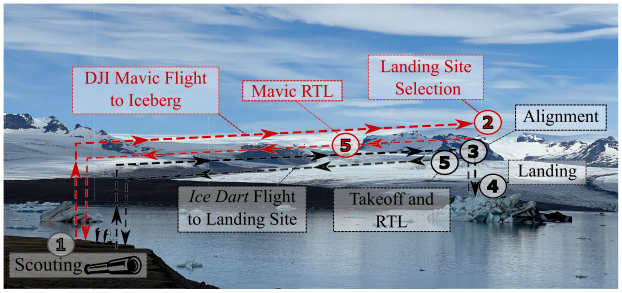


FIGURE 15. **Mission sequence during the field trials, numbered in chronological order. The operations performed by the DJI Mavic are illustrated in red, whereas the operations performed by the *Ice Dart* are illustrated in black.**

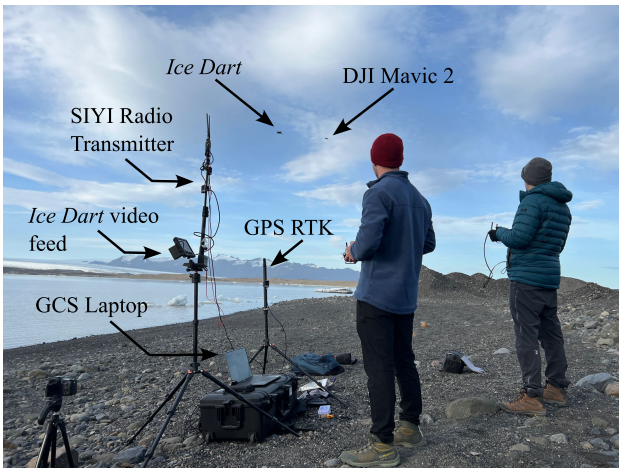


FIGURE 16. **GCS and UAV setup deployed during a mission.**

ground because the suspension was able to fully compress and engage the spines.

To demonstrate that multiple landing sites can be found even on very irregularly shaped icebergs, three landings were conducted on the same large iceberg at different locations (tests 18, 19, and 20). Fig. 18 shows the three landing sites: one on relatively flat ice at 17° (18), another on a steep 46° face (19), and the last at the top of the iceberg (20). Many more sites could have been identified on this iceberg if needed.

Considering that a key goal of the global landing strategy was for the drone to descend in a straight line and brake aggressively after touchdown to prevent sliding and the risk of falling into holes or cracks, Fig. 19 displays the maximum positional error during the descent phase and the final braking distance on the ice. These were recorded from the drone's onboard data. The positional error during descent also provides a better understanding of the drone's stability in strong wind conditions. All tests showed horizontal errors of less than 18 cm between the start of the descent and the initial impact point, indicating a stable descent with minimal influence from winds and gusts. In addition, although tests 9 to 18 lacked RTK corrections and experienced strong winds in some cases, the drone still managed to descend without significantly deviating from the intended landing site. Similarly, Fig. 19 indicates that all tests except two had sliding distances under 19 cm, even in cases with heading offsets as large as 45° with respect to the landing surface. This suggests the feasibility of selecting small landing zones near holes, ice protrusions, cracks, or other ice features that could have endangered the mission. The sliding distance tends to increase with the slope angle, especially above 30°. The two tests with higher sliding distances of 25.9–26.6 cm (tests 7 and 8) can be attributed to the fact that at slope angles above 35°, the descent speed of 1.5 m/s and RT of 60% were typically reduced to 1 m/s and increased to at least 80%, respectively. However, in both cases, the descent speed and RT remained unchanged to assess their impact on landing behavior and performance.

To further demonstrate the *Ice Dart*'s ability to land on icebergs of various shapes and sizes, and to show that a large touchdown envelope simplifies and speeds up the landing site selection process, an experiment was conducted to land on multiple icebergs from the same GCS within a 2-h window. This included scouting icebergs, choosing landing sites with the DJI Mavic, performing and filming the landings, returning to the GCS, swapping batteries every 2–4 flights, and periodically downloading flight logs and footage. This demonstration took place on day two, at the second GCS of the day, on the eastern side of Jökulsárlón lagoon. Fig. 20 shows the resulting iceberg coverage, illustrating the six icebergs within 600 m of the GCS that were successfully landed on. Some icebergs with flat, easily accessible surfaces near the GCS were deliberately skipped to challenge the drone. The results emphasize that the proposed landing strategy allows for efficient operation and extensive coverage, potentially enabling multiple drones to track and monitor drifting icebergs after calving. Overall, the proposed landing solution makes selecting landing sites quick and simple, unlike other methods that might require detailed surface scanning to find suitable landing zones.

C. LESSONS LEARNED

The *Ice Dart* generally performed as expected during the trials. As the team gained experience with mission operations, the difficulty was progressively increased. The drone seemed to land very easily with minimal risks, which helped lower stress levels during testing. The following paragraphs will outline some of the key lessons learned during the trials.

First, one particular element that proved challenging was the yaw alignment of the *Ice Dart* relative to the ice slope. Indeed, judging slope direction from above through a camera on reflective white ice is difficult, as shown in Fig. 21. As a result, most landings were performed with a heading offset relative to the slope gradient, as reported in Table 1. As mentioned in Section IV, heading offsets inherently lead to asymmetric contact, where the legs impact the ice at different timings, rather than following an ideal uphill–downhill contact sequence. Fortunately, the *Ice Dart*'s 30° outward leg stance proved to be a useful design strategy, as it

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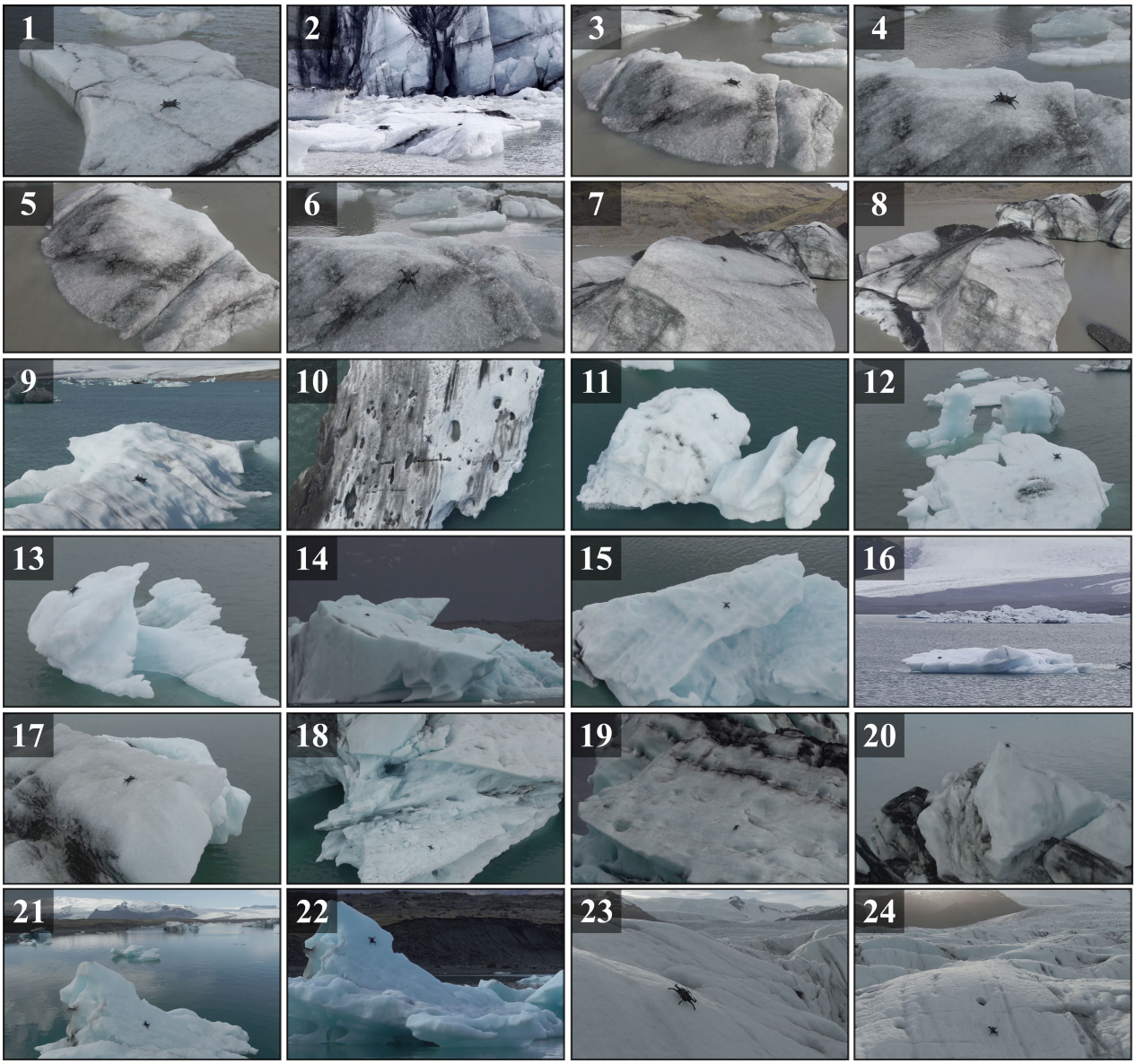


FIGURE 17. Mosaic of landing sites showing all the icebergs and glaciers where the *Ice Dart* landed. Tests 1–8 were done on day 1 of trials, tests 9–17 on day 2, and tests 18–24 on day 3. More information about all the tests can be found in Table 1.

provided a strong tolerance and robustness to yaw offsets, which enabled safe landings even on a 48° slope with a 32° heading error (test 21), or on a 42° slope with a 45° heading error (test 23). This feature greatly simplified and accelerated the landing operation as the drone did not have to be perfectly aligned to perform the landings, helping make the field trials a success.

Second, since cryospheric environments are known for consistently strong winds and gusts, the *Ice Dart* was designed with a high thrust-to-weight ratio of six for improved power and maneuverability, a high-precision GNSS for better control and landing accuracy, and a reduced landing speed setpoint of 1–1.5 m/s to maintain a safety margin in case of sudden velocity changes due to gusts. Although efforts were made to fly during days with less wind, recorded winds remained strong. Contrary to our expectations, winds up to 30 km/h had minimal effects on the drone's descent speed or trajectory. As such, the landing gear, which can theoretically withstand impacts of up to 3.8 m/s, has never encountered such conditions in the field and could be made significantly lighter. Similarly, the thrust-to-weight ratio could be lowered for better efficiency. Both adjustments could extend the mission's range or allow for carrying more payload.

Finally, selecting a landing site and aligning the *Ice Dart* represented the main challenge. Since it's hard to judge the slope angle and alignment from a single perspective, the Mavic had to move to different viewpoints to verify the alignment. This necessitated coordination between the two pilots. The process of choosing the landing site with

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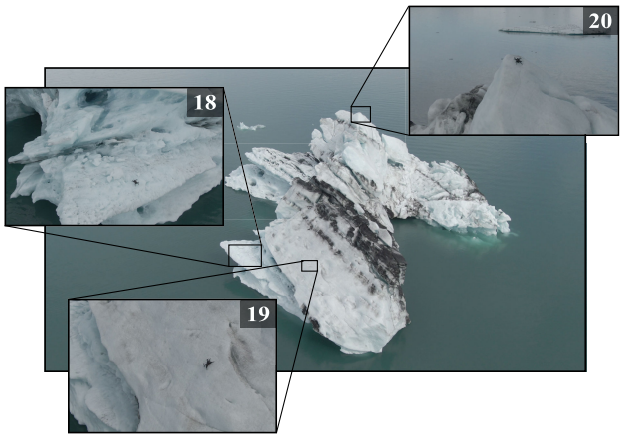


FIGURE 18. Three landings performed at different locations on the same iceberg (tests 18, 19, and 20).

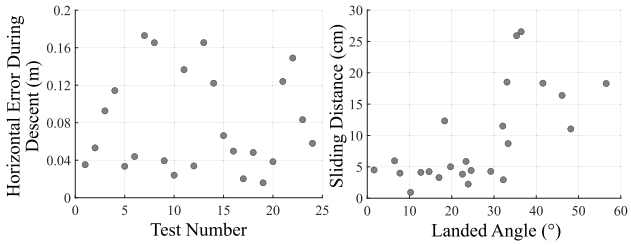


FIGURE 19. Horizontal positional error at impact relative to the start of descent (left). Total sliding distance traveled by the UAV during impact, from first contact to final landed position, as a function of the ice surface angle (right).

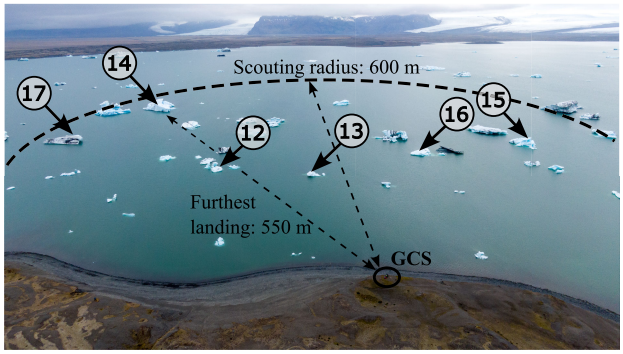


FIGURE 20. Landings completed within a 2-h period on the east side of Jökulsárlón lagoon. Icebergs were scouted in a radius of 600 m. The circled numbers correspond to the test identification numbers used in Fig. 17.

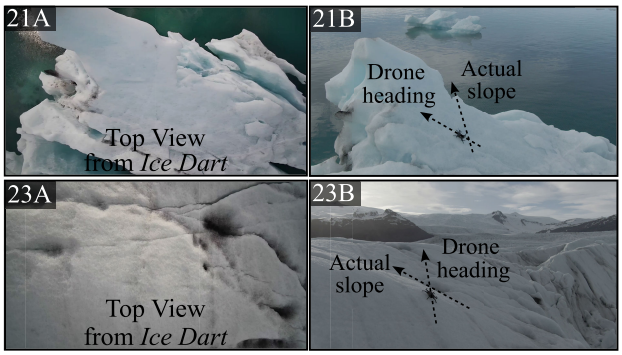


FIGURE 21. Illustration of the misalignment of the drone's heading with respect to the slope. The left column displays the perspective of the *Ice Dart*'s onboard camera, and the right column shows the resulting heading of the drone after landing. The top and bottom rows correspond to tests 21 and 23, respectively, with heading errors of 32° and 45° relative to the maximum gradient of the slope.

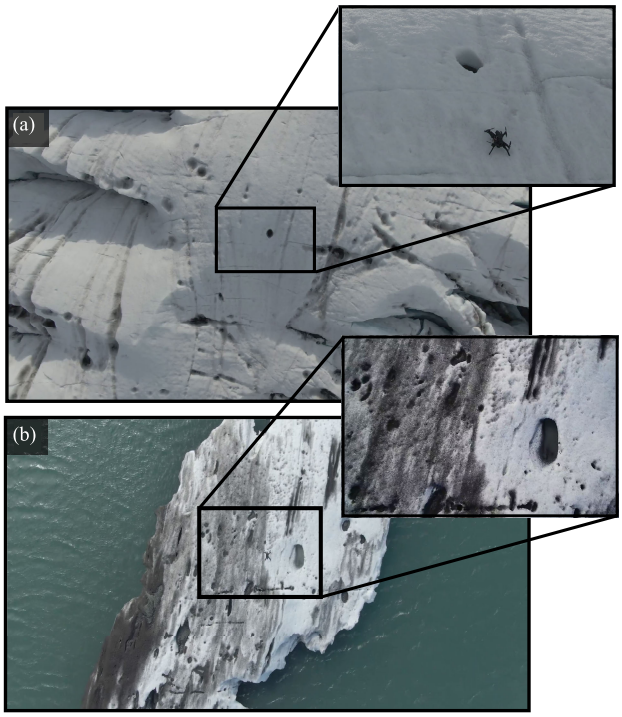


FIGURE 22. Examples of holes and crevasses that are to be avoided when landing on icebergs and glaciers. (a) Test 24, landing on the Fjallsjökull glacier. (b) Test 10, landing on an iceberg in Jökulsárlón.

the Mavic usually took up to 5 min, with an additional 0.5–2 min for alignment (about 5% to 20% of the *Ice Dart*'s flight time). Given the *Ice Dart*'s landing capabilities, identifying suitable sites and aligning the drone could be relatively easy to perform with onboard sensors. These sensors could measure the slope angle and direction, assess the site's flatness, and avoid holes or crevasses, as shown in Fig. 22. This would save time and reduce the pilot's workload, freeing them to focus on the overall mission objectives, such as defining a general area of interest. Such site selection could be done using a lidar or an RGB-D camera.

VII. CONCLUSION

This article introduced a landing system that allows small multirotors to reliably land on steep ice slopes like those on icebergs and glaciers. The system combines three components: 1) a new spiny foot with opposing, passively retractable spines that produce anisotropic shear forces for both uphill- and downhill-facing landings; 2) a friction-based dissipative landing gear that absorbs kinetic energy

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and prevents rebound to enhance spine penetration; and 3) RT at touchdown to increase normal forces and significantly boost shear force generation, especially on steep slopes where gravity alone provides minimal normal force. A study of spine–ice interaction guided the selection of spine diameter, length, and inclination. This led to two configurations: a larger, longer 60° spine for high shear force with RT on the downhill feet, and a smaller 45° spine that reengages easily after slipping on the uphill feet. Both spines are integrated into a single spiny foot, spring-loaded and configured to rotate in opposite directions, enabling passive, self-engagement in the ice depending on the sliding direction. These feet are mounted on a four-legged dissipative landing gear attached to a custom quadrotor platform called the *Ice Dart*. Drop tests at impact speeds of 1–3 m/s on an inclinable ice platform confirmed successful landings on slopes up to 60°, while field trials in Iceland demonstrated 24 successful landings on various ice surfaces (i.e., rough, white, sedimented, or smooth blue ice), at inclinations up to 58°, and impact speeds up to 1.55 m/s. Landings were also achieved at distances up to 554 m, in winds of up to 30 km/h with gusts, and with heading offsets of up to 45°. All landings succeeded, with average braking distances of 9.3 cm and no observed damage or dulling of the spines, demonstrating the system's effectiveness and durability.

Beyond landing on icebergs and glaciers, the proposed approach has broader implications for UAV deployment in hazardous environments. Future tests could target landings on icebergs in high sea states, where icebergs are subject to both horizontal and vertical motion, and where winds are likely to be stronger. This would enable drones to serve as autonomous trackers on smaller, fast-drifting icebergs where installing instruments from helicopters is impractical [10], allowing improved modeling of iceberg drift and continuous monitoring to mitigate collision risks with offshore structures. Preliminary tests have also shown that the spines perform well on hard surfaces such as wood or shingles, suggesting that a less aggressive version of the spiny feet could improve landing performance on other rough or inclined terrains. Future work will likely focus on enhancing system maturity through automated landing-site detection and alignment, aiming to reduce pilot workload. Since field trials confirmed stable descent even in strong winds, efforts will also aim to reduce the landing gear's weight to extend flight time, increase payload capacity, and enable longer iceberg-tracking missions.

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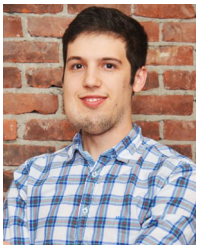
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